Joshua Dinn

joshdinn01@gmail.com | +61 429 556 122 | linkedin.com/in/josh-dinn | Portfolio: dinnyyy.github.io

Education

Monash University, Melbourne, Australia **Mar 2023 – Oct 2027**

Bachelor of Engineering (Software Engineering) and Bachelor of Commerce (Econometrics)

WAM: 79.97 (Distinction) | GPA: 3.48 / 4.0 | 100% in FIT2107 (Software Quality & Testing)

Experience

Collingwood Football Club, Data, Analytics and Technology Intern **Mar 2026 – Jun 2026**

- Built a membership churn prediction model in XGBoost, engineering feature pipelines from historical transaction and engagement data to help commercial stakeholders identify at-risk members.
- Built an interactive dashboard visualising churn drivers and SHAP feature importance, giving commercial stakeholders a self-serve view into retention risk.
- Designed and implemented a repository synchronisation workflow between the club's web and mobile codebases, removing a manual step from every deployment.

Monash eSolutions, AI Fitness Trainer **Jul 2026 – Present**

- Run in-person sessions teaching students and staff how to use AI effectively, ethically, and in everyday study and work.
- Assist in delivering a university webinar on AI to an audience of 200+ attendees.

Monash Deep Neuron, Optimised Computing Member **Sep 2025 – Present**

- Building the on-device AI memory system for AR glasses: a vector database with HNSW indexing and approximate nearest-neighbour (ANN) search powering a retrieval-augmented generation (RAG) pipeline that grounds responses in prior context with low latency.
- Teaching Assistant for an 8-week C++ and Accelerated Computing workshop on GPU programming, CUDA, and HPC, run with NVIDIA, Pawsey Supercomputing Centre, and NCI.
- NVIDIA CUDA certified.

Outlier AI, AI Trainer **Jun 2024 – Dec 2025**

- Evaluated AI-generated technical solutions written in LaTeX against correctness, logic, and formatting benchmarks, building a consistent rubric-driven review process.
- Delivered structured feedback on model reasoning and output quality that directly informed iterative improvements to model performance.

Projects

Distributed Vector Database (C++20) **Ongoing**

- Building a high-performance vector database from scratch in C++20, hand-optimising SIMD (AVX2) distance kernels and a custom memory allocator for cache-efficient nearest-neighbour search at scale.
- Extending the system with a learned query planner (gradient-boosted trees) and a full RAG pipeline on top of the index, aimed at dynamically trading off recall against latency instead of relying on fixed parameters.

StdOut: Real-Time AI Technical Interview Simulator **Feb 2026**

- Built a real-time interview simulation platform with a 3-person hackathon team using React, Node.js, MongoDB, and WebSockets.
- Engineered a low-latency pipeline combining live transcription, AI-driven conversation, and in-browser coding, handling streaming updates asynchronously so they never blocked the session.
- Designed the system architecture to synchronise candidate and interviewer state in real time over WebSockets, keeping transcription, AI responses, and code changes consistent throughout the session.
- Built a Python code-execution terminal with event-based tracking of code diffs and session timelines, enabling full replay of a candidate's coding process.

Interactive Maze Solver & Algorithm Visualiser **Oct 2025**

- Built a pathfinding platform visualising 7 algorithms across procedurally generated mazes.
- Added automated benchmarking and metrics tracking across 125,000+ simulated maze runs.

Algorithmic Trading System for S&P 500 ETF **May 2025**

- Built and backtested a trading system using Hidden Markov Models (HMMs) to classify market regimes for SPY.
- Integrated RSI, ATR, and volatility signals into position sizing and trade logic.
- Deployed on an AWS Linux server with the Alpaca API for automated paper trading.

Skills & Interests

Languages: Python, C++, JavaScript, Java, MATLAB, SQL, R

Econometrics: Maximum likelihood estimation, panel data and fixed effects models, cross-sectional methods, time-series modelling (GARCH, HMM)

Frameworks & Tools: React, PyTorch, MongoDB, AWS, Linux, Git, pandas, NumPy

Interests: System design, machine learning, statistical modelling